Introduction of VR

3D Stereo Vision & Haptic

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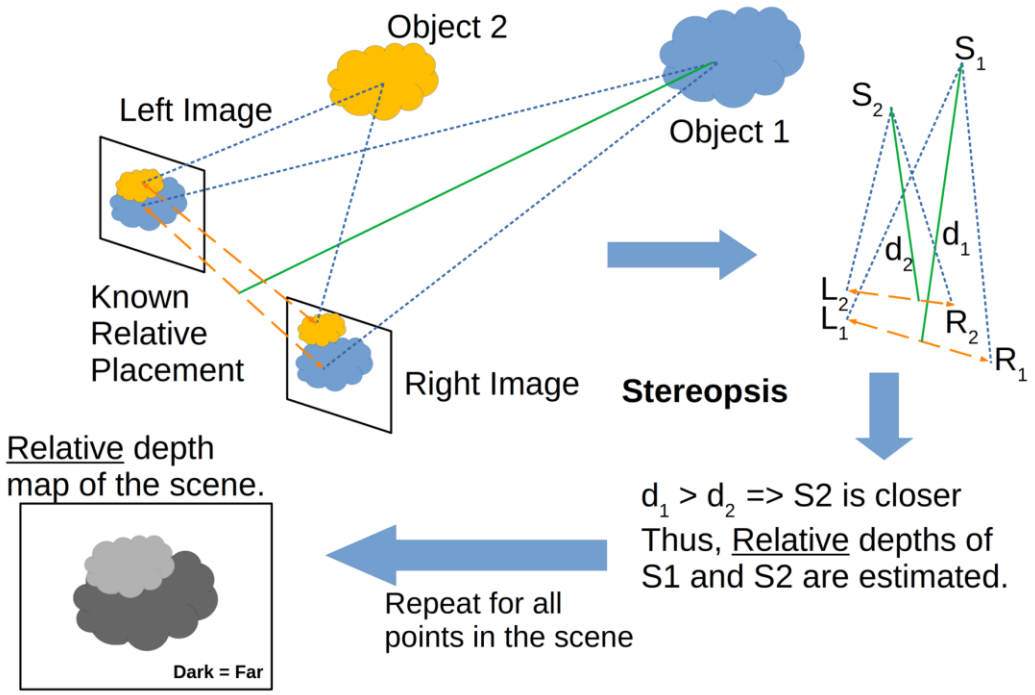
3D Stereo Vision

- As with other types of 3D imaging technology, the goal with stereo vision is to solve the issue of depth perception – the core of all 3D problems.
- Stereo vision is a machine vision technique that can provide full field of view 3D measurements using two or more machine vision cameras.
- The foundation of stereo vision is similar to 3D perception in human vision and is based on triangulation of rays from multiple viewpoints.

3D Stereo Vision

- A stereo camera closely copies how our eyes work to give us accurate, real-time depth perception.
- It achieves this by using two sensors a set distance apart to triangulate similar pixels from both 2D planes.

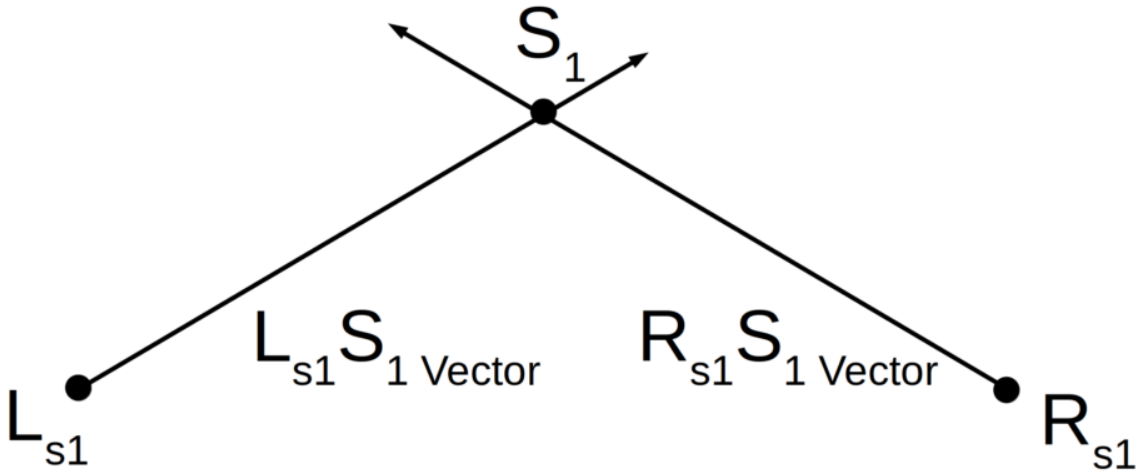
- Each pixel in a digital camera image collects light that reaches the camera along a 3D ray.
- If a feature in the world can be identified as a pixel location in an image, we know that this feature lies on the 3D ray associated with that pixel.



3D Stereo Vision

- If we use multiple cameras, we can obtain multiple rays.
- A stereo pair of images will thus have direction vectors emanating from the 2D pixels representing a common 3D point in the 3D scene. All points on a direction vector are candidate sources. Since two vectors can intersect at only one unique point, we take the intersection point as the source:

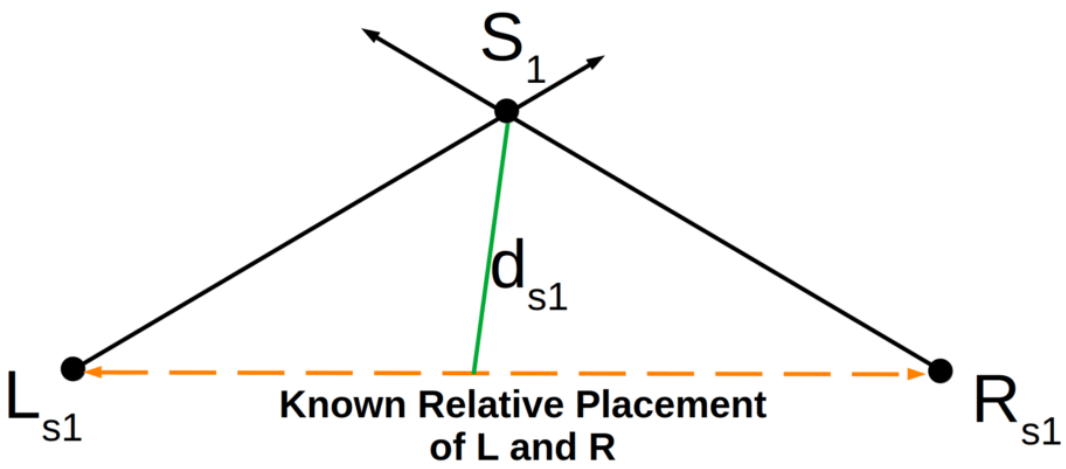
The direction vectors from the left and right images ($L_{S1}S_1$ ad $R_{S2}S_1$, respectively) intersect at the single source. This 3D source point in the scene is the point from where light rays cast image pixels



3D Stereo Vision

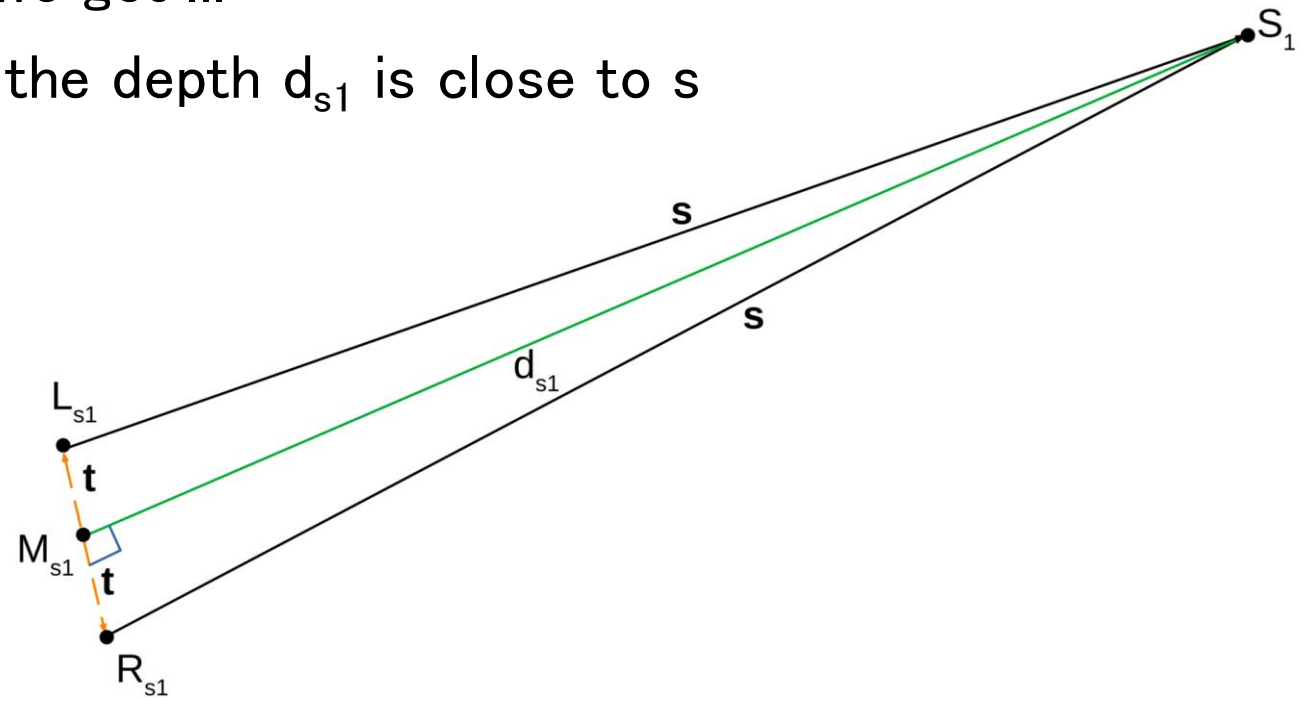
- We assume that we know the distance between cameras and that it's very small compared to the distance between the object and the cameras. Under that assumption, we can determine the location of the 3D point in space by triangulation.

1. The actual depth ds1 for the point from the line joining the two cameras.
2. Let's note that the angle between the line ds1 and the lne LS1RS1 is not exactly 90 degrees
3. The distance LS1RS1 is very small compared to dS1
4. This results in the angle between the line dS1 and the line LS1RS1 being approximately 90 degrees
5. We determine the location of S1 by triangulation, and we know the relative distance LS1RS1, we can calculate the depth dS1 using Pythagorean theorem.



3D Stereo Vision

- Since s is very large compared to t, the angle $S_1M_{S1}R_{S1}$ aproaches 90.
- Lengths $L_{S1}M_{S1}$ and $M_{S1}R_{S1}$ are almost the same (denoted y t)
- Lengths $L_{S1}S_1$ and $R_{S1}S_1$ are almost the same (denoted by s)
- Applying the Pythagorean theorem, we get ...
- Since s is very large compared to t, the depth d_{s1} is close to s



Stereo Vision

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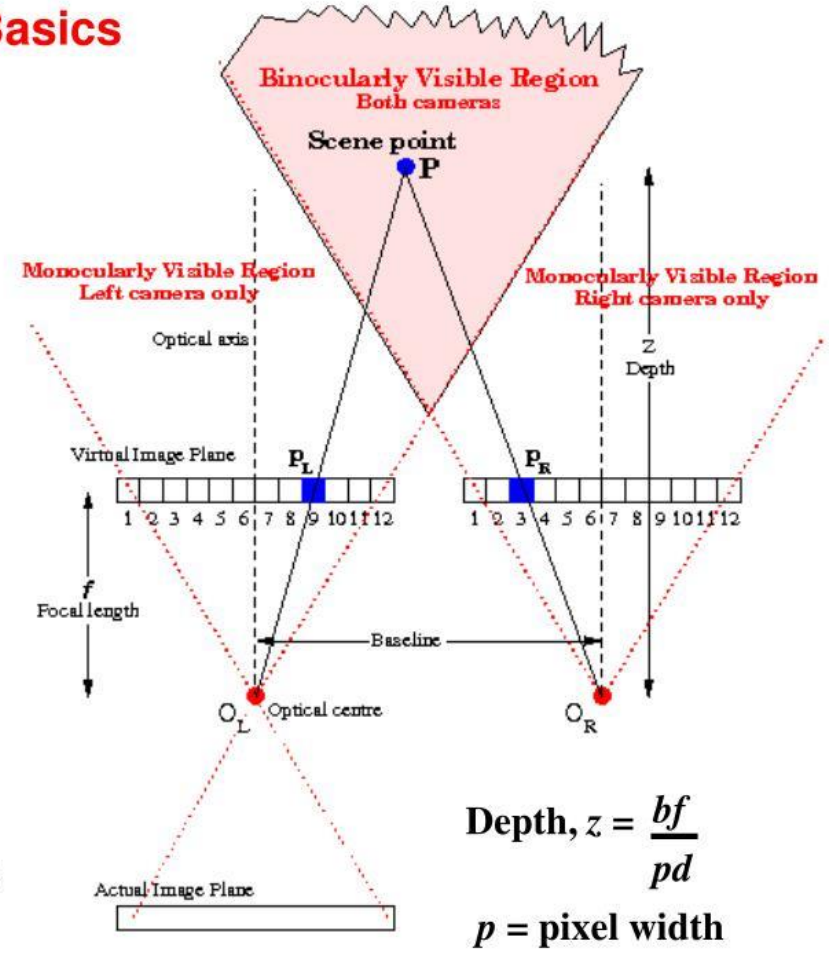
Stereo Vision - Basics

Two cameras: Left and Right
Optical centres: O_L and O_R
Virtual image plane is projection of actual image plane through optical centre
Baseline, b, is the separation between the optical centres
Scene Point, P, imaged at p_L and p_R

$p_L = 9$
$p_R = 3$
Disparity, $d = p_R - P_L = 6$

Disparity is the amount by which the two images of P are displaced relative to each other



Depth, $z = \frac{bf}{pd}$

p = pixel width

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Haptics

- We interact with a virtual environment using a plurality of portable devices that mainly affect our visual and aural perception.
- Formed from five basic senses (sight, hearing, touch, smell, taste)
- The main senses in the perception of reality are vision and hearing
- These two embrace about 90% of the day-to-day experience of the person without perceptual disorders and the honorable third place with approximately 10% goes to the sense of touch
- Every day we interact with a ton of electronic devices, some of them we carry around (like laptops or mobile phones) with other being wearable (like smart watches, wristbands, clothing)

Haptic

- Haptics is a technology, which allows one to receive tactile information through their sensations, by applying forces, vibrations, or touches. Haptics simulates an object or interaction from the virtual system, producing the feeling it's real.
- Mobile phone vibration is very often described as an example of haptic feedback technology. But it is just one and very simple illustration of how haptics can function.
- Haptics allows a user to interact with computer-based devices by receiving tactile and force feedback. The former may let us know what the texture of the object is (e.g. rough or smooth). The latter simulates some physical properties of the object, such as its weight or pressure.

Haptic feedback typology

- Five main types of haptic feedback technologies (haptics) are
  - force,
  - vibrotactile,
  - electrotactile,
  - ultrasonic, and
  - thermal.

Force feedback

- Force feedback stimulates the ligaments and muscles through our skin into the musculoskeletal system, whereas any other types of haptics affect top layers of skin receptors
- The cutaneous devices (involving the outer layer of the skin) are quite compact and apply the acupressure on small areas of the body.
- In contrast, force devices are mostly large (think about a powered exoskeleton as an example). They move together with a human and have an impact on large areas of the body, such as an arm or a leg. These devices are far more complex, as they are designed to both apply the force to the body part, and provide a person with sufficient freedom of movement.

Force feedback

- There are two types of force feedback devices by human body parts emulation: biomimetic and non-biomimetic
  - Biomimetic devices move as human limbs and resemble them by their forms
  - Non-biomimetic devices may be very different from the human body.

- Another classification of the force feedback appliances (by the direction of the applied power) includes resistive and active devices
  - Resistive devices limit the movement of the user with the help of the brakes
  - Active devices restrict movements of the user or move the body in space by means of motors

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Vibrotactile Feedback

- Vibrotactile feedback is by far the most common type of haptics.
- Vibrostimulators apply pressure to the definite receptors of human skin. These receptors resemble “onion” layers’ structure and can accept vibrations of up to 1000 hertz.
- Ordinary human speech frequency varies from 80 to 250 hertz, so our skin can actually feel sounds.

Vibrotactile Feedback

Actuator Type:

- **ERM (Eccentric Rotating Mass)**: Simple and affordable, provides basic vibrations but limited frequency range and directionality.
- **LRA (Linear Resonant Actuator)**: More precise and controlled vibrations, higher frequency range, often used for directional feedback.
- **ECAT (Electrocutaneous Actuation)**: Simulates tingling sensations directly on the skin, not widely used due to potential discomfort.
- **Ultrahaptics**: Uses focused ultrasound to create mid-air haptic sensations, still experimental but holds promise for future applications.

Performance:

- **Frequency Response**: Range of frequencies the device can output (20-500 Hz for basic textures, up to 1 kHz for finer details).
- **Amplitude**: Intensity of vibrations (measured in g-force or displacement).
- **Number of Actuators**: Single for basic feedback, multiple for complex sensations and localized effects.
- **Resolution & Dynamic Range**: Ability to create subtle variations in intensity and frequency for nuanced feedback.

Vibrotactile Feedback

Other Considerations:

- **Form Factor:** Size, weight, and placement of the device on the body or object.
- **DurabPower Consumption:** Important for battery-powered VR headsets.
- **ility:** Ability to withstand repeated use and potential abuse.
- **Latency:** Time delay between virtual event and haptic feedback (ideally under 10 ms for immersion).
- **Safety:** Skin compatibility and compliance with safety regulations.

Additional Standards:

- **VT-1 Specification:** Defines performance parameters for high-quality, realistic vibrotactile feedback.

Examples of existing devices:

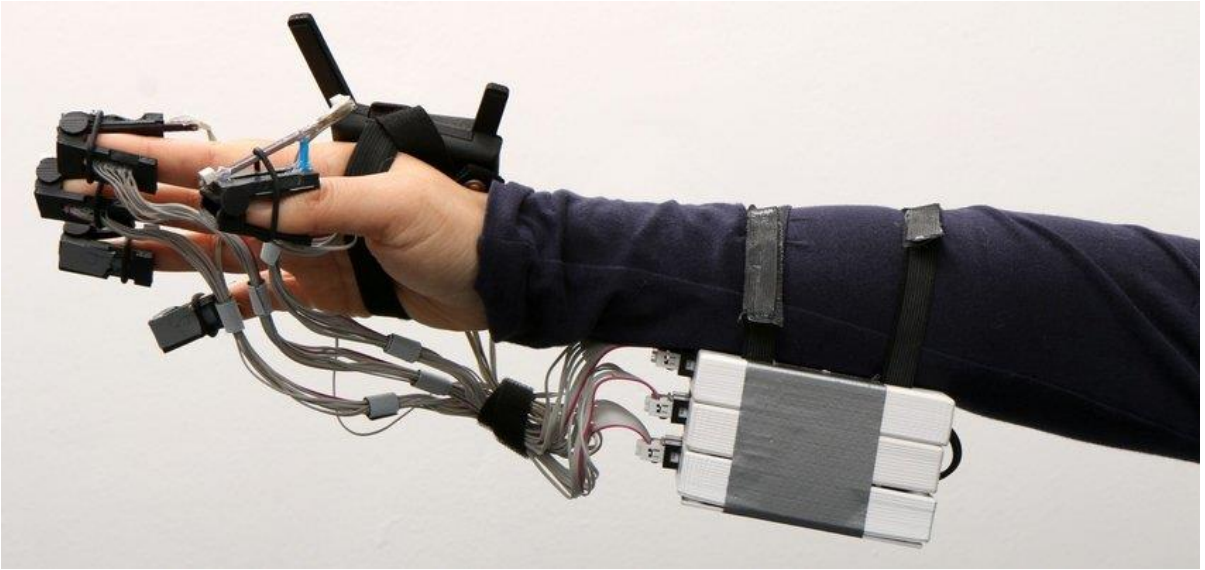
- **Oculus Touch Controllers:** Use ERM for basic vibrations.
- **Valve Index Controllers:** Use LRAs for more precise haptic feedback.
- **Tesla Suit:** Uses multiple types of actuators to provide full-body haptic feedback.

Electrotactile Feedback

- Electrotactile stimulators affect both receptors and nerve endings by applying electrical impulses. By means of electrical impulses, a user can receive a wide range of sensations which can not be reproduced with any other current feedback systems.
- This type of feedback has many forms depending on the intensity and frequency of the stimuli delivered to the skin.
- Sensations can also vary depending on the current, voltage, material, wave form, electrode size, contact force, hydration, and skin type.

Electrotactile Feedback

- The principal advantage of the electro-haptic feedback system, compared to vibrotactile or force feedback, is the absence of mechanical or moving parts.
- Another benefit of electro-neural stimulation is that the electrodes can be assembled into compact arrays and used to implement electrotactile displays.



Ultrasonic Tactile Feedback

Technology:

- Utilizes focused ultrasound waves to create mid-air pressure variations simulating touch sensations on the user's skin.
- Different implementations exist, including phased arrays and acoustic levitation, each with varying characteristics.

Performance:

- **Resolution:** Determines the size and detail of perceivable haptic points. Current devices offer millimeter-scale resolution, with potential for further improvement.
- **Intensity:** Controls the perceived "strength" of the touch sensation. Ranges are still under development, aiming to match and surpass existing haptic methods.
- **Focus and Directivity:** Ability to precisely deliver sensations to specific points on the user's hand or body. Current devices show promising directivity, with ongoing research for refined focusing.
- **Frequency Range:** Affects the perceived texture and roughness of virtual objects. Early devices operate in the 40-100 kHz range, targeting skin sensitivity.

Ultrasonic Tactile Feedback

Other Considerations:

- **Form Factor:** Devices currently range from bulky setups to wearable bracelets, with miniaturization being an active research area.
- **Safety:** Ultrasound intensity needs to be within safe limits for prolonged skin exposure. Current research focuses on optimizing this balance.
- **Latency:** Achieving low latency (ideally under 10 ms) is crucial for immersion. Existing systems are still under development in this aspect.
- **Cost:** Currently, UTF technology is at the research stage, making devices expensive. Cost reduction is expected with further development and adoption.

Thermal Feedback

- For thermal feedback formation, the actuators grid is used. It is in direct contact with the skin. Most commonly, to implement this effect, the thermoelectric diodes (based on the Peltier effect) are used.
- People don't define well the place of the thermal stimulus unlike with tactile communication. Therefore, there is no need in many actuators to create a heat or cold feedback, and they can be positioned not so close to each other. Thus, in some way, thermal feedback devices are even easier to design.

Thermal Feedback

Technology: Various approaches exist, including:

- **Thermoelectric Devices (TEDs):** Produce heating and cooling effects by converting electricity to temperature changes.
- **Peltier Modules:** Similar to TEDs but offer higher efficiency and faster response times.
- **Microfluidic Systems:** Use circulating liquids to achieve precise temperature control.
- **Infrared (IR) Lasers:** Focus heat on specific skin areas for highly localized effects.

Performance:

- Temperature Range: Ability to heat and cool within a safe and comfortable range (typically 25-45° C).
- Response Time: Speed of changing temperature (ideally under 100 ms for realistic simulation).
- Spatial Resolution: Number of independent heating/cooling elements for localized feedback (varies depending on technology).
- Intensity Control: Precisely adjust the perceived warmth or coolness.
- Power Consumption: Important for battery-powered VR applications.

Thermal Feedback

Other Considerations:

- **Form Factor:** Size, weight, and placement of the device on the body or object (flexibility is crucial for wearability).
- **Skin Contact:** Material and design need to ensure safe and comfortable skin contact (consideration for allergies and irritation).
- **Safety:** Temperature control measures to prevent skin burns or discomfort.
- **Durability:** Ability to withstand repeated use and potential abuse.
- **Latency:** Time delay between virtual event and thermal feedback (similar to other haptic methods, ideally under 10 ms).

Examples:

- **TEGway ThermoReal:** Uses Peltier modules for heating and cooling in VR gloves and controllers.
- **SenseGlove Touch:** Microfluidic system for thermal feedback in haptic gloves.
- **VRgluv**: Combines vibration and thermal feedback in a VR glove.

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Assignment

- Describe detailed techniques used in stereo vision to measure the depth dimension.
- Advantages and disadvantages of stereo vision
- Describe these following techniques
  - 3D calibration
  - 3D matching
  - 3D registration
- List all haptic modalities, their techniques and applications
- List and describe some common navigation techniques in VR

VR Quiz

We define VR as the experience "in which the user is effectively immersed in a responsive virtual world. This implies user dynamic control of viewpoint." Who gave this definition in which year?	a) Ivan Sutherland, 1968 b) Jaron Lanier, 1989 c) Fred Brooks, 1999 d) Luckey Palmer, 2012
Which of the following feature is NOT part of the difference between immersive VR and other media?	a) User dynamic control of viewpoint via head-tracking b) 3D stereo vision c) Surrounding d) Resolution
What does VR (for instance, in an Oculus HMD) add to the experience in comparison to a 3D cinema experience? (Select all that apply)	a) 3D stereovision b) In a 3D cinema, everybody sees the same thing on display. In HMD VR, the images are updated in real time according to the user's viewpoint. c) HMD VR provides a surrounding experience where there is no escape.

VR Quiz

Generally speaking, what is the difference between the CAVE and the HMD?	a) The CAVE is a multiple user device while the HMD is for one single user. b) The HMD provides 3D stereovision, the CAVE doesn't. c) The CAVE takes a large space and is normally setup in research labs. The HMD is more portable.
Generally speaking, which of the statement is true about mobile VR?	a) Mobile VR has lower resolution b) Mobile VR has no rotation tracking c) Mobile VR is less equipped at supporting 3D interaction
Which of the following is NOT always part of items to consider when purchasing VR headsets?	a) Resolution b) Tracking c) GPU (Graphic Processing Unit) d) Comfort
Who built the first digital VR system, and in which year?	a) Ivan Sutherland, 1968 b) Jaron Lanier, 1989 c) Fred Brooks, 1999

VR Quiz

Not all VR devices come with position tracking – some has rotation tracking only. Why is position tracking beneficial? (Select all that apply)	a) It allows you to rotate your head and look around in VR. b) It allows you to move your body in a naturalistic way to observe an object up close, or look at table from underneath. c) It allows you to explore the environment with your body naturally, for instance you can walk around the environment and see things from different angles.
We often use VR controllers to: (Select all that apply)	a) Interact with objects b) Gesture in a social communication c) Navigate the environment
If you would like to develop a VR game aiming for the mass market, which device should you choose?	a) High-end HMDs b) Mobile VR c) The CAVE
What's the difference between the refresh rate of a VR device and the framerate of the experience it delivers?	a) They are the same b) Refresh rate could be a lot higher than the framerate you end up having in your application. c) Refresh rate could be a lot lower than the framerate you end up having in your application.

VR Quiz

Which of the followings statement is true about the weight and comfort of VR HMDs?	a) The lighter the more comfortable. b) Mobile VR is always lighter than high-end HMDs c) High-end HMDs can be lighter than mobile VR
Which of the following technical features are true when comparing the CAVE and the HMD? (Select all that apply)	a) In the CAVE you can have lower framerate. b) The CAVE has a bigger field-of-view c) The HMD provides a better surrounding experience than the CAVE
The CAVE was invented by:	a) Carolina Cruz-Neira, 1992 b) Jaron Lanier, 1989 c) David Swapp, 2004
How could we track position of the head without external camera/sensors?	a) We cannot do that. b) Use inside-out tracking c) Use inertial tracking

Projects

Projects	Yêu cầu
Quá trình (cá nhân) (tuần 12-13)	Vẽ object 3D trên Blender Tiêu chí đánh giá: • Đẹp và đồ sộ • Độ khó • Có tính sáng tạo • Chấm điểm tại chỗ (kêu vẽ lại tại chỗ)
Cuối kỳ (nhóm 2 người) (tuần 13-14)	Thiết kế và hoàn thành game 3D trên Unity Tiêu chí đánh giá: • Phải là 3D • Game có tính logic • Có các chế độ chơi khác nhau • Có độ khó nhất định • Có hệ thống đánh giá điểm thưởng và qua màn • Có tính sáng tạo trong gameplay • Có hiệu ứng đồ họa đẹp (ví dụ như cháy nổ)